Introduction to Machine Learning
Unit 6 Solutions: Logistic Regression

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1. Suggest possible response variables and predictors for the following classification problems. For each problem, indicate how many classes there are. There is no single correct answer.

   (a) Given an audio sample, to detect the gender of the voice.

   (b) A electronic writing pad records motion of a stylus and it is desired to determine which letter or number was written. Assume a segmentation algorithm is already run which indicates very reliably the beginning and end time of the writing of each character.

   Solution:

   (a) Target is the gender (M or F), so there are two classes. One predictor could be the vector of digital samples of the audio. Later, we will discuss some other popular features extracted from these samples.

   (b) Target is the character written (e.g. 0-9, a-z and A-Z). So, there would be 26+26+10 classes plus classes for punctuation. The predictor could be a vector of (x, y) positions of the stylus. Each position may also include a pressure. Of course, the details would depend on the stylus interface implementation.

2. Suppose that a logistic regression model for a binary class label $y = 0, 1$ is given by

$$P(y = 1|\mathbf{x}) = \frac{1}{1 + e^{-z}}, \quad z = \beta_0 + \beta_1 x_1 + \beta_2 x_2,$$

   where $\boldsymbol{\beta} = [1, 2, 3]^\mathsf{T}$. Describe the following sets:

   (a) The set of $\mathbf{x}$ such that $P(y = 1|\mathbf{x}) > P(y = 0|\mathbf{x})$.

   (b) The set of $\mathbf{x}$ such that $P(y = 1|\mathbf{x}) > 0.8$.

   (c) The set of x_1 such that $P(y = 1|\mathbf{x}) > 0.8$ and $x_2 = 0.5$.

   Solution:

   (a) Note that

$$P(y = 0|\mathbf{x}) = 1 - P(y = 0|\mathbf{x}) = 1 - \frac{1}{1 + e^{-z}} = \frac{e^{-z}}{1 + e^{-z}}.$$

Hence,

$$P(y=1|\mathbf{x}) > P(y=0|\mathbf{x}) \iff \frac{1}{1+e^{-z}} > \frac{e^{-z}}{1+e^{-z}} \iff e^{-z} < 1$$
$$\iff z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 > 0.$$

(b) In this case,

$$P(y=1|\mathbf{x}) > 0.8 \iff \frac{1}{1+e^{-z}} > 0.8$$
$$\iff z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 > -\ln\left[\frac{0.2}{0.8}\right] = \ln(4).$$

(c) Combining with part (b), the set is,

$$1 + 2x_1 + 3x_2 > \ln(4) \text{ and } x_2 = 0.5 \Rightarrow x_1 > \frac{\ln(4) - 2.5}{3}.$$

3. A data scientist is hired by a political candidate to predict who will donate money. The data scientist decides to use two predictors for each possible donor:

- x_1 = the income of the person (in thousands of dollars), and
- x_2 = the number of websites with similar political views as the candidate the person follow on Facebook.

To train the model, the scientist tries to solicit donations from a randomly selected subset of people and records who donates or not. She obtains the following data:

Income (thousands \$), x_{i1}	30	50	70	80	100
Num websites, x_{i2}	0	1	1	2	1
Donate (1=yes or 0=no), y_i	0	1	0	1	1

Solution:

(a) The scatter plot is shown in Fig. 1.

(b) A simple classifier is to use the boundary $x_2 = 0.5$, shown on the figure. So, we take

$$z_i = x_2 - 0.5 = \mathbf{w}^\mathsf{T}\mathbf{x} + b,$$

where $\mathbf{w} = [0, 1]$ and $b = -0.5$.

(c) We have

$$P(y_i = 1|\mathbf{x}_i) = \frac{1}{1+e^{-z_i}} \Rightarrow P(y_i = 0|\mathbf{x}_i) = 1 - \frac{1}{1+e^{-z_i}} = \frac{1}{1+e^{z_i}}.$$

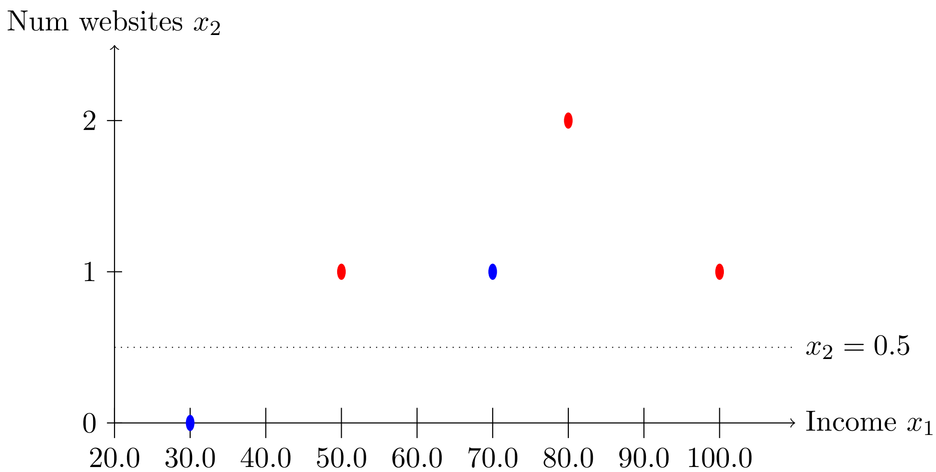

Figure 1: Scatter plot of the data points where the red circles are $y_i = 1$ and blue are $y_i = 0$

Hence, we can write

$$P(y_i|\mathbf{x}_i) = \frac{1}{1+e^{-u_i}}, \quad u_i = \begin{cases} z_i & \text{if } y_i = 1, \\ -z_i & \text{if } y_i = 0 \end{cases}$$

Since $1/(1+e^{-u})$ is increasing in u, the likelihood will be minimized for the sample where u_i is the smallest. We calculate u_i for each sample using the following table:

Income (thousands \$), x_{i1}	30	50	70	80	100
Num websites, x_{i2}	0	1	1	2	1
Donate (1=yes or 0=no), y_i	0	1	0	1	1
$z_i = x_{i2} - 0.5$	-0.5	0.5	0.5	1.5	0.5
u_i	0.5	0.5	-0.5	1.5	0.5

We see u_i is smallest for sample $i = 3$, which is the misclassified point.

(d) Let z_i' be the new values of the linear discriminant under the new parameters. We have,

$$z_i' = (\mathbf{w}')^{\mathsf{T}}\mathbf{x}_i + b' = \alpha \left[\mathbf{w}^{\mathsf{T}}\mathbf{x}_i + b\right] = \alpha z_i.$$

Since $\alpha > 0$, the sign of z_i' is the same as z_i. Therefore $\hat{y}_i$ does not change. However, the probabilities do change. Since $\alpha > 1$,

$$z_i > 0 \Rightarrow z_i' > z_i$$
$$z_i < 0 \Rightarrow z_i' < z_i.$$

Hence, for samples where $P(y_i = 1|\mathbf{x}_i) > 0.5$, the probability will increase. For samples where $P(y_i = 1|\mathbf{x}_i) < 0.5$, the probability will decrease.

(e) You can use the following code:

```python
import numpy as np
def gen_rand(X,w,b):
    z = X.dot(w)+b[:,None]
    p = 1/(1+np.exp(-z))
    n = X.shape[0]
    u = np.random.rand(n)
    y = (u < p)
    return y
```

4. Suppose we collect data for a group of students in a machine learning class with variables X_1 = hours studied, X_2 = undergrad GPA, and Y = receive an A. We fit a logistic regression and produce estimated coefficient, $\beta_0 = -6,\ \beta_1 = 0.05,\ \beta_2 = 1$.

   (a) Estimate the probability that a student who studies for 40 h and has an undergrad GPA of 3.5 gets an A in the class.

   (b) How many hours would the student in part (a) need to study to have a 50 % chance of getting an A in the class?

Solution:

(a) The linear discriminator is:

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 = 6 + 0.05(40) + (1)(3.5) = -0.5,$$

so the probability is:

$$P(Y = 1|\boldsymbol{X}) = \frac{1}{1+e^{-z}} \approx 0.38.$$

(b) For $P(Y = 1|\boldsymbol{X}) = 0.5$, we need $z = 0$ so:

$$\begin{aligned} z &= \beta_0 + \beta_1 X_1 + \beta_2 X_2 \\ \iff X_2 &= \frac{1}{\beta_1}\left[z - \beta_0 - \beta_1 X_1\right] = \frac{1}{0.05}\left[0 + 6 - (1)(3.5)\right] = 50 \text{ h}. \end{aligned}$$

5. The loss function for logistic regression for binary classification is the binary cross entropy defined as

$$J(\boldsymbol{\beta}) = \sum_{i=1}^{N} \ln(1 + e^{z_i}) - y_i z_i$$

where $z_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i}$ for two features $x_{1,i}$ and $x_{2,i}$.

   (a) What are the partial derivatives of z_i with respect to β_0, β_1, and β_2.

   (b) Compute the partial derivatives of $J(\boldsymbol{\beta})$ with respect to β_0, β_1, and β_2. You should use the chain rule of differentiation.

   (c) Can you find the close form expressions for the optimal parameters $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\beta}_2$ by putting the derivatives of $J(\boldsymbol{\beta})$ to 0? What methods can be used to optimize the loss function $J(\boldsymbol{\beta})$?

Solution:

(a) The partial derivatives are:

$$\frac{\partial z_i}{\partial \beta_0} = 1, \quad \frac{\partial z_i}{\partial \beta_1} = x_{i1}, \quad \frac{\partial z_i}{\partial \beta_2} = x_{i2}.$$

(b) The partial derivative of J with respect to z_i is:

$$\frac{\partial J}{\partial z_i} = \frac{e^{z_i}}{1 + e^{z_i}} - y_i,$$

so using chain rule:

$$\frac{\partial J}{\partial \beta_j} = \sum_i \frac{\partial J}{\partial z_i} \frac{\partial z_i}{\partial \beta_j} = \sum_i \left[\frac{e^{z_i}}{1 + e^{z_i}} - y_i \right] \frac{\partial z_i}{\partial \beta_j},$$

where you can substitute the expression for $\frac{\partial z_i}{\partial \beta_j}$ from part (a).

(c) There is no closed form solution, but you can use any numeerical optimizer such as gradient descent.